EFU METALLURGY LICENCE v1.0

Experimental EFU-based Steel, Aluminum, and Metal Processing Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for metallurgy (steel, aluminum, copper, zinc, lead).

NOT a furnace permit, air quality, or slag disposal license replacement, but an open research protocol measuring metallurgy's physical footprint in human-scale EFU metrics (coke, ore, slag, CO_2). Covers primary (ore-based), secondary (scrap), and electrolysis operations.

I. Core Concepts

1. **EFU-Metallurgy (EFU-METAL)**
   EFU framework application for metal production:
   - **EFU-E_METAL**: coke, electricity, fuel oil (furnace/melting)
   - **EFU-F_METAL**: iron ore, bauxite, scrap (feedstock)
   - **EFU-EM_METAL**: CO_2, CO, dioxins, heavy metal dust
   - **EFU-S_METAL**: slag, mill scale, galvanizing baths
   - **EFU-W_METAL**: cooling water, gas scrubbing
2. **Metallurgy Sovereignty Gap (SS_METAL)**
   SS_METAL = TEFU_METAL_total – Regional Metallurgy Carrying Capacity (RMCC_METAL)
   Measures if production stays within regional ore/scrap regeneration capacity.
3. **MROI_METAL**
   MROI_METAL = THI_METAL / TEFU_METAL
   THI_METAL = construction steel + automotive parts + recycling rate.

II. EFU Metallurgy Licence – Research Status

Documents EFU-audited steelworks (ArcelorMittal, Alcoa).
Does NOT replace blast furnace permits, emissions trading, or waste permits.
Enables primary (ore) vs. secondary (scrap) metallurgy benchmarking.

III. Metrics and Thresholds

1. EFU Components (per tonne finished steel/metal)

EFU-E_METAL (Energy)
EFU-E_METAL = energy_MWh / 12.88 / t_steel
Example: Primary steel = 20 MWh/t → 0.00155 EFU/t

EFU-F_METAL (Feedstock)
EFU-F_METAL = ore_scrap_t / 300kg/person/year / t_steel
Example: 1.6t ore/t steel → 0.0053 EFU/t

TEFU_METAL = 1.4×EFU-E + 1.6×EFU-F + 2.0×EFU-EM + 1.3×EFU-S + 1.2×EFU-W

2. Regional Metallurgy Carrying Capacity (RMCC_METAL)

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RMCC_METAL = Sustainable_steel_Mt / Population / 1t/person/year
Example: China = 1,000Mt / 1,410M = 0.71 EFU_METAL/person/year
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IV. Licence Categories (EFU-METAL Labels)

Level	SS_METAL	MROI_METAL	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-METAL SOVEREIGN**	Scrap-based
2	0 < ≤25k EFU	≥5e-3	**EFU-METAL BALANCED**	Mixed feedstock
3	25k < ≤125k EFU	≥1e-3	**EFU-METAL DEFICIT**	Ore-dependent
4	>125k EFU	<1e-3	**EFU-METAL CRITICAL**	Furnace limits

V. Regional RMCC_METAL Calibrations

Country	Steel (Mt/year)	Population (M)	RMCC_METAL (EFU/person/year)
CHINA	1,000	1,410	0.71
INDIA	120	1,400	0.009
USA	80	340	0.02
EU	150	448	0.03
JAPAN	90	125	0.07

VI. Example: Chinese Steel Industry (Baowu)

Input (annual sector):

- Steel: 1,000Mt
- Energy: 20,000B MWh
- Ore: 1,600Mt
- Slag: 2,000Mt

- Workers: 2M staff
- Region: China (RMCC_METAL = 0.71 EFU/person/year)

Calculation:

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EFU-E_METAL = 20T/12.88/1T = 0.00155 EFU/t
EFU-F_METAL = 1.6T/300/1T = 0.0053 EFU/t

TEFU_METAL_total ≈ 850M EFU/year
Allocated = 2M × 0.71 = 1.42M EFU/year
SS_METAL = 850M – 1.42M = +848M EFU/year (Level 4: CRITICAL)

Result: EFU-METAL CRITICAL
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Sub-sector Comparison: Electric arc furnace = 0.3 EFU/t (Level 2: BALANCED)

VII. Implementation Notes

1. **EU CBAM**: TEFU_METAL basis for steel import tariffs
2. **Green steel**: EFU-F_METAL metric for scrap content verification
3. **Data sources**: Tier 1 = coke receipts; Tier 2 = World Steel Association